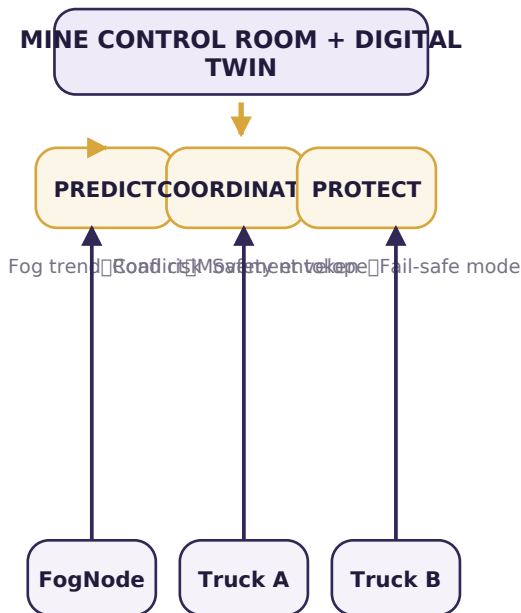


FOGNET NEXUS

Predictive Cooperative Safety Network for Open-Pit Mine Haul Roads



The upgrade: FogNet does not only detect fog. It predicts risky road conditions, predicts truck-to-truck conflicts, coordinates who moves and who waits, and changes to a safer operating mode when sensors or communication fail.

SIH26007 • Easy-to-explain architecture • Prototype first • Production path clearly separated

1. What problem are we solving?

In an open-pit mine, trucks can enter fog, meet another truck around a blind curve, lose communication, or approach a dangerous road edge. A camera-only system reacts after the danger is already visible. FogNet Nexus tries to know the danger earlier.

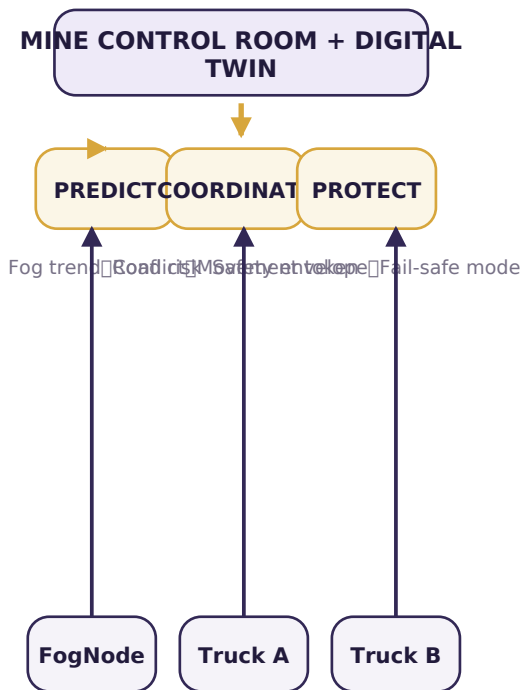
The new idea in one sentence

Make the mine road cooperative. Every truck and roadside node shares useful safety information, while each truck can still protect itself when the network is unavailable.

Five jobs

1	Predict	Detect rising fog/visibility risk before a truck enters the area.
2	See	Combine local sensors to understand obstacles and road conditions.
3	Coordinate	Predict truck conflicts and give one vehicle a temporary movement token.
4	Protect	Calculate a safe speed and safety state continuously.
5	Degrade safely	If communication or a sensor fails, reduce the operating envelope instead of silently continuing.

2. Complete FogNet Nexus architecture



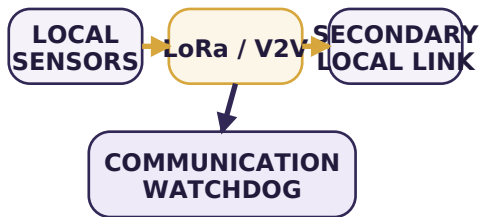
Important: the control room helps coordinate the fleet, but a truck is not made helpless if the network disappears. Local sensors remain the first protection layer.

Main building blocks

FogNode	Visibility/light scatter + temperature/humidity + LoRa. Gives early warning for a road segment.
Truck safety unit	Camera, ultrasonic for prototype, GPS/IMU, LoRa, buzzer/display and local safety logic.
Safety Core	Risk calculation, stopping margin, safe-speed recommendation, conflict prediction and operating state.
Traffic Coordinator	Predicts whether two truck safety envelopes may overlap and creates a movement token.
Control room	Mine map, vehicle status, fog-risk map, communication health, alerts and safety black box.

3. What if LoRa fails?

This is a major judge question. The answer should not be 'we have another radio'. The stronger answer is: communication quality changes the truck's safety state.



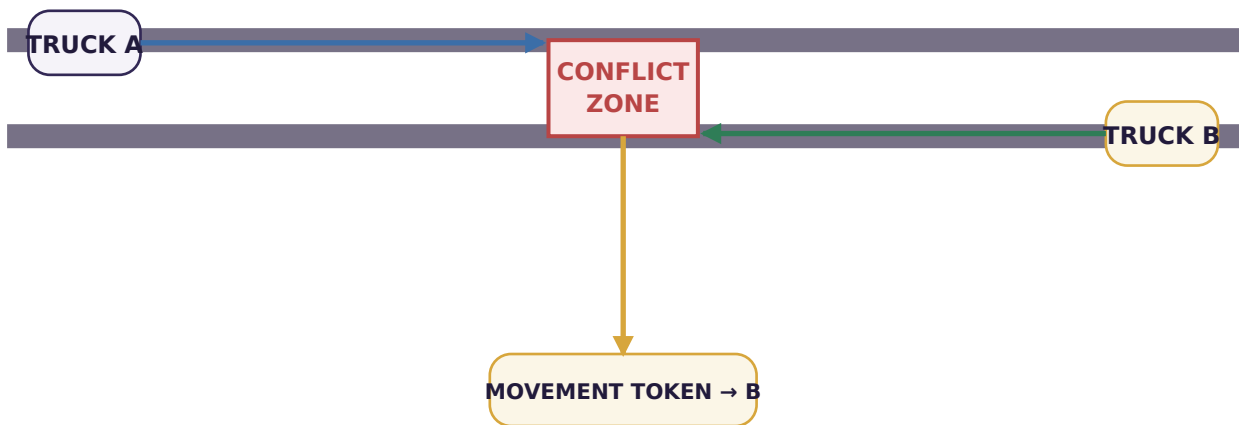
Communication logic

GREEN	Good link	Normal operation.
YELLOW	Some loss / delay	Increase safety margin and reduce permitted speed.
RED	Link lost	Local-only safety mode. Local sensors become the main protection. Supervisor is informed when a link returns.

Key rule: communication is supplementary. A truck must never assume that an old V2V message is current. Every message carries a timestamp/age and sequence number. Stale information is rejected.

4. Loaded truck vs unloaded truck: who gets priority?

Do not hard-code 'loaded truck always wins'. That can create a new risk. Instead, FogNet calculates which movement is safer based on the actual road situation.



Dynamic Right-of-Way Engine

Vehicle	Loaded/unloaded state, current speed, stopping ability
Road	Width, gradient, blind curve, passing bay and road geometry
Environment	Visibility/fog level and current hazard level
Position	Distance to conflict zone and whether the truck has already entered it
Safety	Which choice gives the larger stopping margin and lower conflict risk

Movement Token: when two paths can conflict, FogNet temporarily gives the road segment to one truck. The other truck receives HOLD / WAIT. After the first truck clears the segment, the token is released.

5. Dynamic Safety Envelope

A fixed rule such as 'warn at 10 metres' is not equally safe at every speed. FogNet compares the distance available with the distance needed to stop.



Fog ↑ | Conflict ↑ | Sensor confidence ↓ | Communication ↓

Simple physics behind the decision

Stopping distance needed = speed × reaction time + speed² / (2 × deceleration)

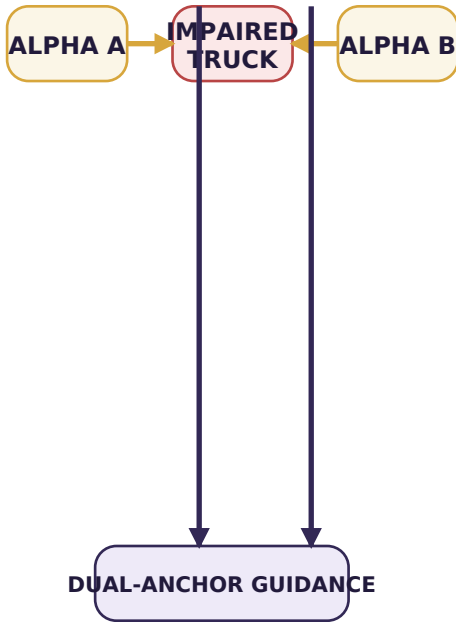
If the usable distance is smaller than the stopping distance needed, the system moves to a more restrictive safety state. The output is a recommended maximum speed and risk state, not only an alarm.

Risk inputs

Visibility	Lower visibility → lower permitted speed.
Obstacle distance	Less distance → higher risk.
Vehicle speed	Higher speed → longer stopping distance.
Road grade	Downhill movement can require more stopping distance.
V2V information	Useful extra information when another truck cannot be seen.
Communication health	Poor communication → larger safety margin.

6. Alpha Mode: temporary safety anchors

Alpha-capable trucks should not behave differently all the time. During normal operation they are ordinary fleet vehicles. If another truck has a serious sensor failure and a safe formation is possible, two anchor-capable trucks can temporarily provide front and rear reference information.



Temporary role • Driver remains in control

Important rule

The software gives an advisory. It does not steer the truck into formation. Drivers remain in control. When the impaired truck's sensors recover, Alpha Mode ends.

One universal dashboard

All trucks should use the same basic driver interface. Alpha capability appears as an extra mode only when activated. This reduces driver confusion and keeps the interface consistent.

7. Failure handling: the system gets safer, not silent

Failure	What FogNet does	Driver / supervisor sees
Camera fails	Reduce trust in camera, enter degraded mode, reduce speed.	SENSOR FAILURE
LoRa fails	Use local safety layer; restrict operation according to risk.	COMMUNICATION LOST
FogNode fails	Predictive warning unavailable for that segment; trucks continue local protection.	NODE OFFLINE
GPS weak	Use IMU for short gaps; reduce confidence for map-based functions.	POSITION DEGRADED
V2V data stale	Reject old packet instead of using it as current.	STALE V2V
Supervisor STOP	Highest-priority safety command.	EMERGENCY STOP

Safety principle: a failure must never make the system more aggressive. It can only keep the same protection or move toward a more conservative operating state.

8. Secure messages + Safety Black Box

Because trucks can receive important commands over radio, messages should not be blindly trusted. For the prototype, use a simple authenticated message format. Production systems would use a stronger security architecture.

Message contains	Vehicle ID • timestamp • sequence number • message type • position • speed • heading • risk state
Receiver checks	Trusted sender • fresh timestamp • valid sequence • valid authentication
If invalid	Reject the message and continue with local safety logic.

Safety Black Box

Every important event is recorded with time and reason: fog risk, V2V detection, conflict prediction, movement-token decision, sensor failure, communication failure, emergency stop and recovery. This lets a supervisor later ask: 'Why did the system tell this truck to stop?'

9. Mine-wide Risk Map and Safe Throughput

FogNet should not stop the whole mine whenever one road segment becomes dangerous. It divides the haul road into segments and calculates a local risk state.



FogNode Conflict Token Failure Recovery

predicts danger before it reaches sight-of-way so safe operation returns

Example

Segment 1 can stay GREEN while Segment 3 is RED. Trucks that are far from the dangerous segment can continue safely, while trucks approaching Segment 3 receive a lower speed or a hold instruction.

Goal: not 'stop every truck'. The goal is maximum safe throughput: keep safe road segments moving while controlling only the traffic that enters the risk zone.

10. The SIH Grand-Finale Demo

The prototype should prove the architecture in a sequence that judges can understand immediately.

0:00	Normal	Two trucks move. Dashboard is green.
0:30	Predict	FogNode sees rising scatter/fog trend. Road segment becomes amber/red before trucks arrive.
1:15	Conflict	Two trucks approach inside fog. Cameras cannot reliably see each other.
2:00	Coordinate	FogNet predicts the conflict and grants a movement token to one truck. The other receives HOLD.
3:00	Communication failure	Disconnect LoRa. Truck enters communication-degraded/local-safety mode.
3:45	Sensor failure	Disconnect camera. Truck enters sensor-degraded mode and reduces permitted speed.
4:30	Alpha Mode	Show temporary anchor guidance as a controlled state transition.
5:15	Emergency	Supervisor sends highest-priority STOP command.
5:45	Black box	Show the complete event history and explain why each action happened.

11. What we build now vs later

Prototype / SIH	Production upgrade
ESP32 + LoRa	Industrial radio/V2X architecture after site testing
Camera + ultrasonic	Camera + mmWave radar + thermal where justified
Physics-informed fog trend	Long-term site data + validated prediction model
GPS/IMU hazard map	Surveyed map + higher-accuracy positioning + local perception
Small demo motor cutoff	OEM-integrated braking/intervention after safety engineering and certification
2 vehicles	Fleet-scale message scheduling and field validation

Claims discipline: do not say the prototype is production-ready. Say what you measured in the demo, and clearly label field-scale capabilities as the next validation stage.

12. Judge-ready answers

What if LoRa fails?

LoRa is not the only safety layer. The truck continues local sensing, communication health changes the safety state, and the vehicle becomes more conservative when the link is degraded or lost.

Why should a loaded truck get priority?

Load is only one factor. FogNet considers road geometry, stopping distance, speed, position, gradient and safe holding locations. The goal is the safest movement, not a permanent 'loaded truck wins' rule.

Why Alpha trucks?

They are not permanently special. Alpha-capable vehicles can temporarily provide front/rear reference data when another truck has a critical sensor failure.

What if GPS is wrong?

GPS is not treated as the only protection against a road edge. Geofencing is a complementary warning layer; production would combine surveyed maps, higher-accuracy positioning and local perception.

Can you automatically brake a real mine truck?

Not with the student prototype. The demo can show a small motor cutoff. Real vehicle intervention needs OEM integration and formal safety engineering.

Is the fog prediction really AI?

For the prototype, call it a physics-informed trend model. Do not claim a trained model has learned real mine fog dynamics unless that has been validated with data.

What makes FogNet different?

The strongest combination is predictive fog risk + cooperative conflict prediction + movement tokens + dynamic safety envelope + graceful failure handling.

Final pitch: “FogNet Nexus makes the mine road cooperative. It predicts dangerous visibility, predicts truck conflicts before they become collisions, coordinates safe movement, continuously calculates a dynamic safety envelope, and becomes more conservative when its sensors or communication fail.”

Source basis: the team's uploaded FogNet solution documents, including the existing layered sensing, predictive FogNode, V2V, dynamic safe-speed, degraded mode, Alpha Mode, geofencing and supervisor-broadcast architecture. This PDF adds the proposed extensions: communication-health states, dynamic movement tokens, cooperative conflict prediction, secure/stale-message handling, and mine-segment throughput coordination.